

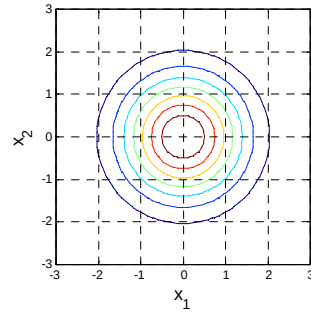
Bivariate (gaussian) distribution, (X_1, X_2) ($p = 2$)

$$\begin{aligned}\mu &= [0, 0] \\ [\sigma_1, \sigma_2] &= [1, 1] \\ \rho &= 0 \\ \Sigma &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\end{aligned}$$

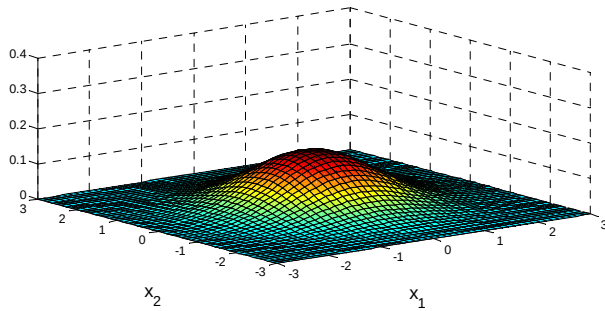
$$\text{total variation} = \sum_i \lambda_{i,\Sigma} = \sigma_1^2 + \sigma_2^2 = 2$$

$$\text{generalized variance} = |\Sigma| = \prod_i \lambda_{i,\Sigma} = \sigma_1^2 * \sigma_2^2 * (1 - \rho^2) = 1$$

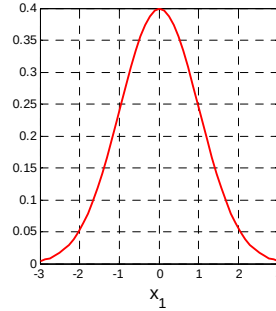
Probability density contour plot for $f(x_1, x_2)$



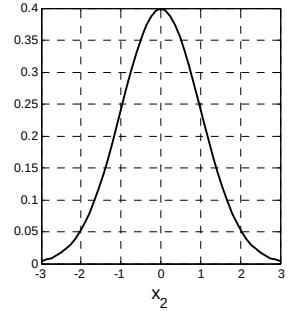
3D plot of simultaneous probability density function $f(x_1, x_2)$



marginal density function $f(x_1)$



marginal density function $f(x_2)$



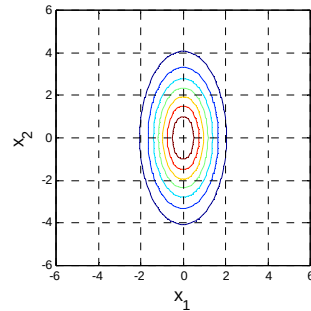
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$$\begin{aligned}\mu &= [0, 0] \\ [\sigma_1, \sigma_2] &= [1, 2] \\ \rho &= 0 \\ \Sigma &= \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}\end{aligned}$$

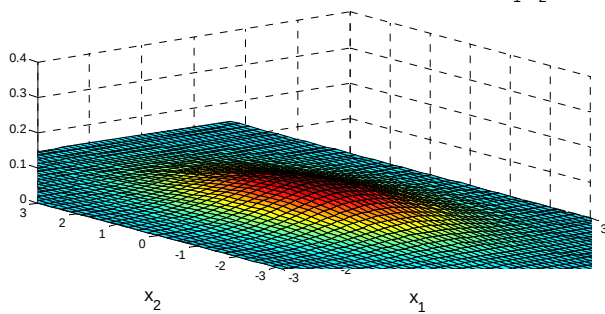
$$\text{total variation} = \sum_i \lambda_{i,\Sigma} = \sigma_1^2 + \sigma_2^2 = 5$$

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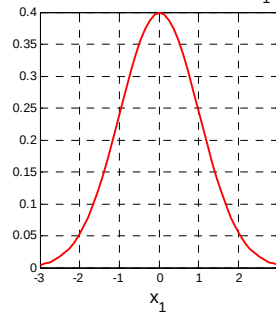
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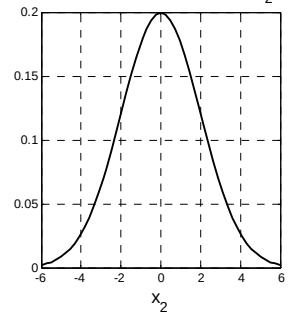
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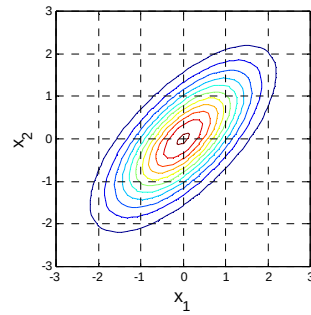
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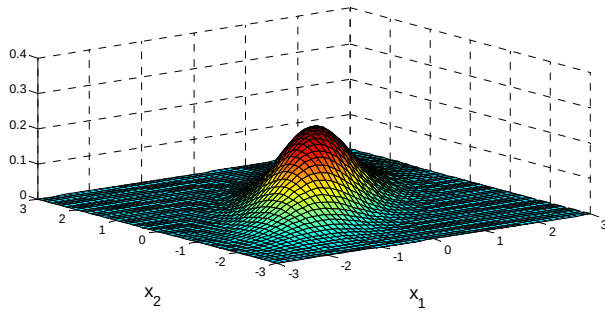
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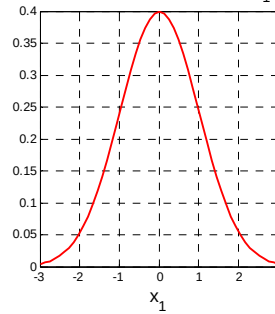
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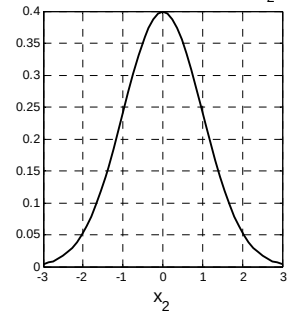
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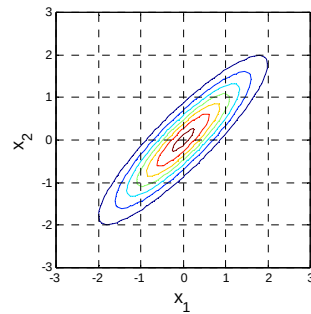
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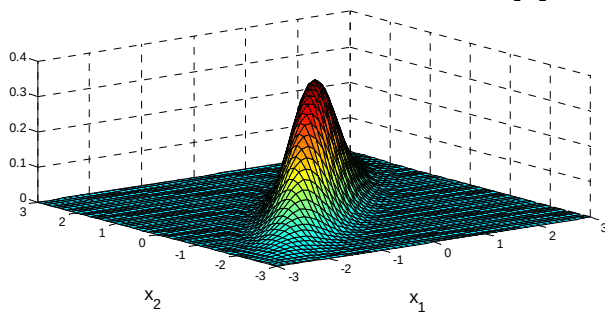
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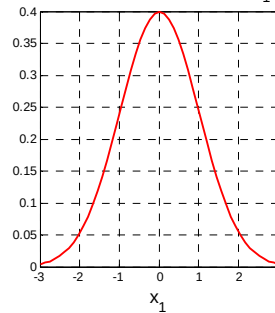
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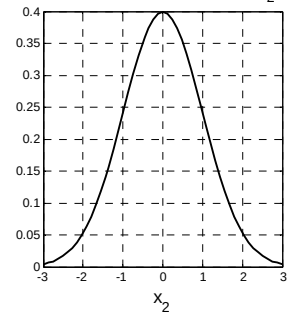
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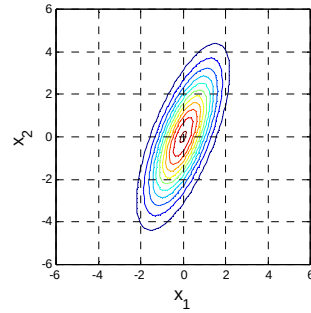
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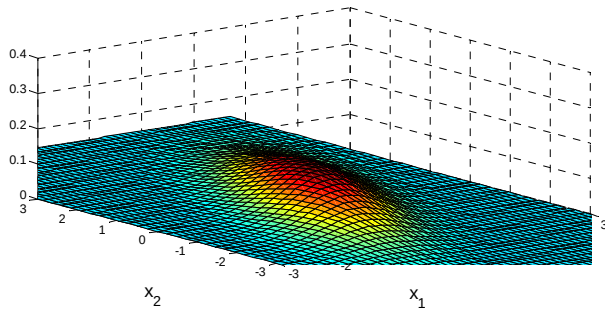
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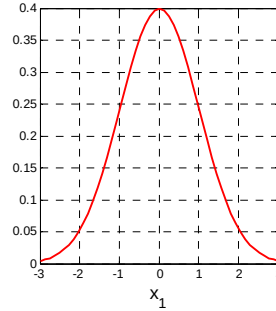
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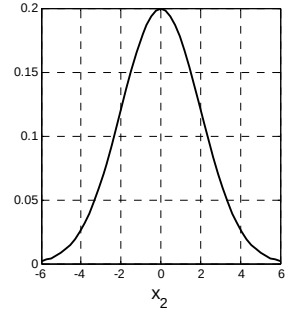
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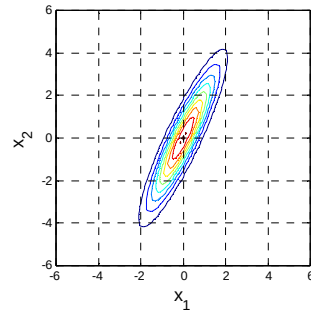
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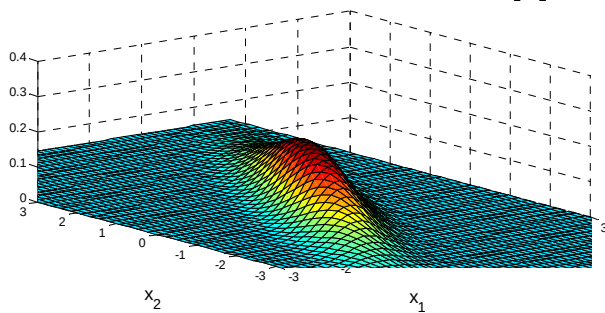
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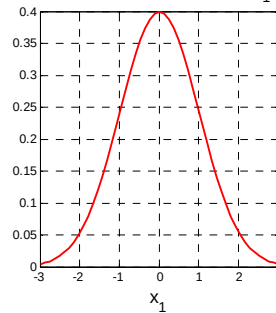
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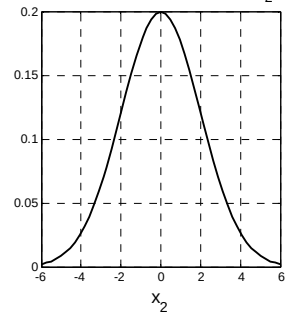
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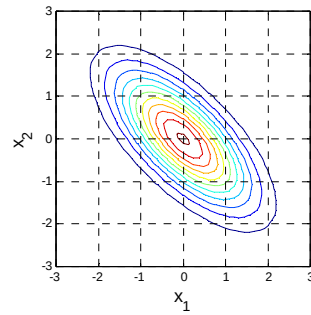
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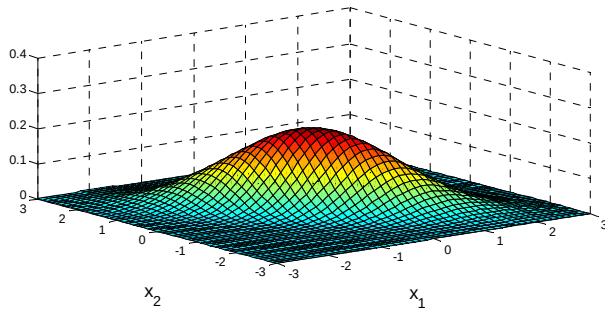
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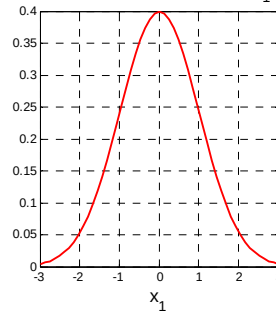
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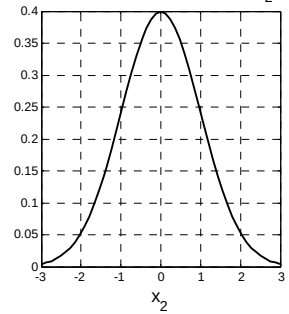
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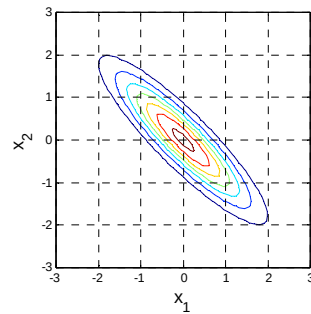
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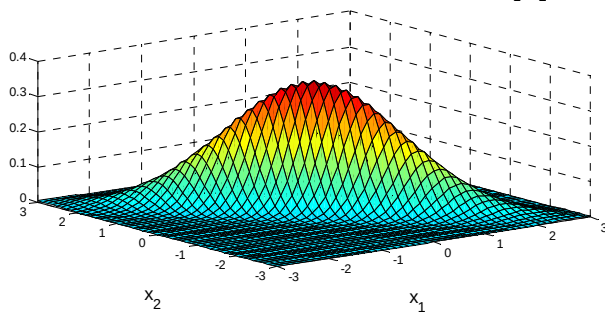
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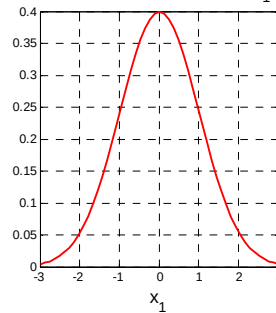
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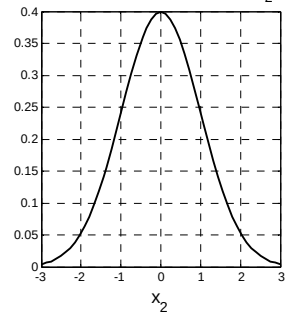
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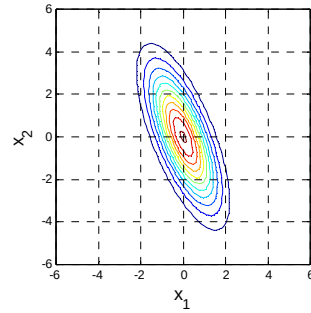
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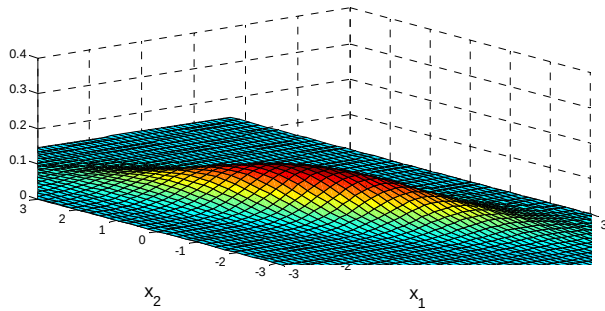
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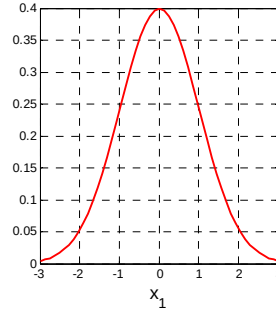
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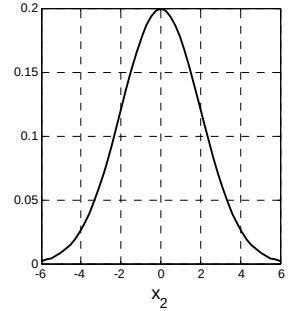
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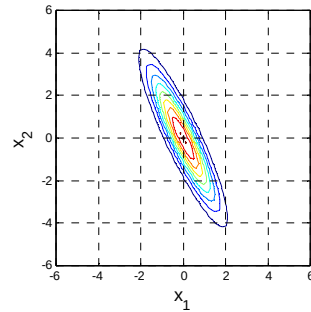
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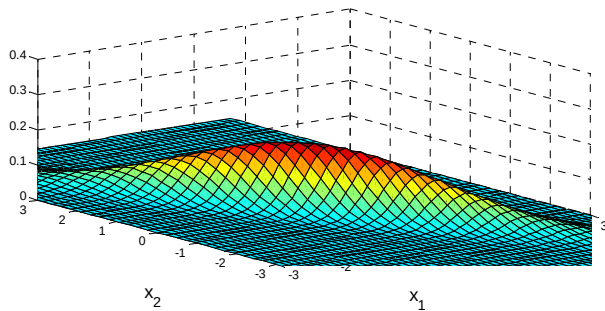
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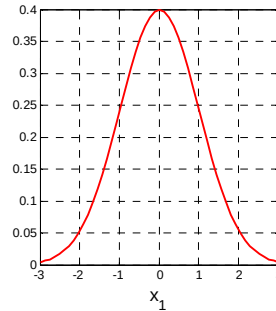
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